

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 28

- PRECIPITATION AND ITS VARIOUS FORMS
- TYPES OF PRECIPITATION-1

PRECIPITATION

1 The term precipitation refers to falling of water, snow or hail from the clouds and it results when condensation is occurring rapidly within a cloud.

2 Although condensation is necessary for precipitation to occur, just the beginning of condensation is not enough to lead to precipitation.

3 If the condensation occurs in very small amounts, the water droplets or ice crystals thus formed may remain floating in air and may be eventually absorbed again by air and no precipitation may occur.

4 Therefore, for precipitation to take place it is necessary that condensation occurs rapidly and continues for sufficiently long time.

PRECIPITATION

FORMS OF PRECIPITATION

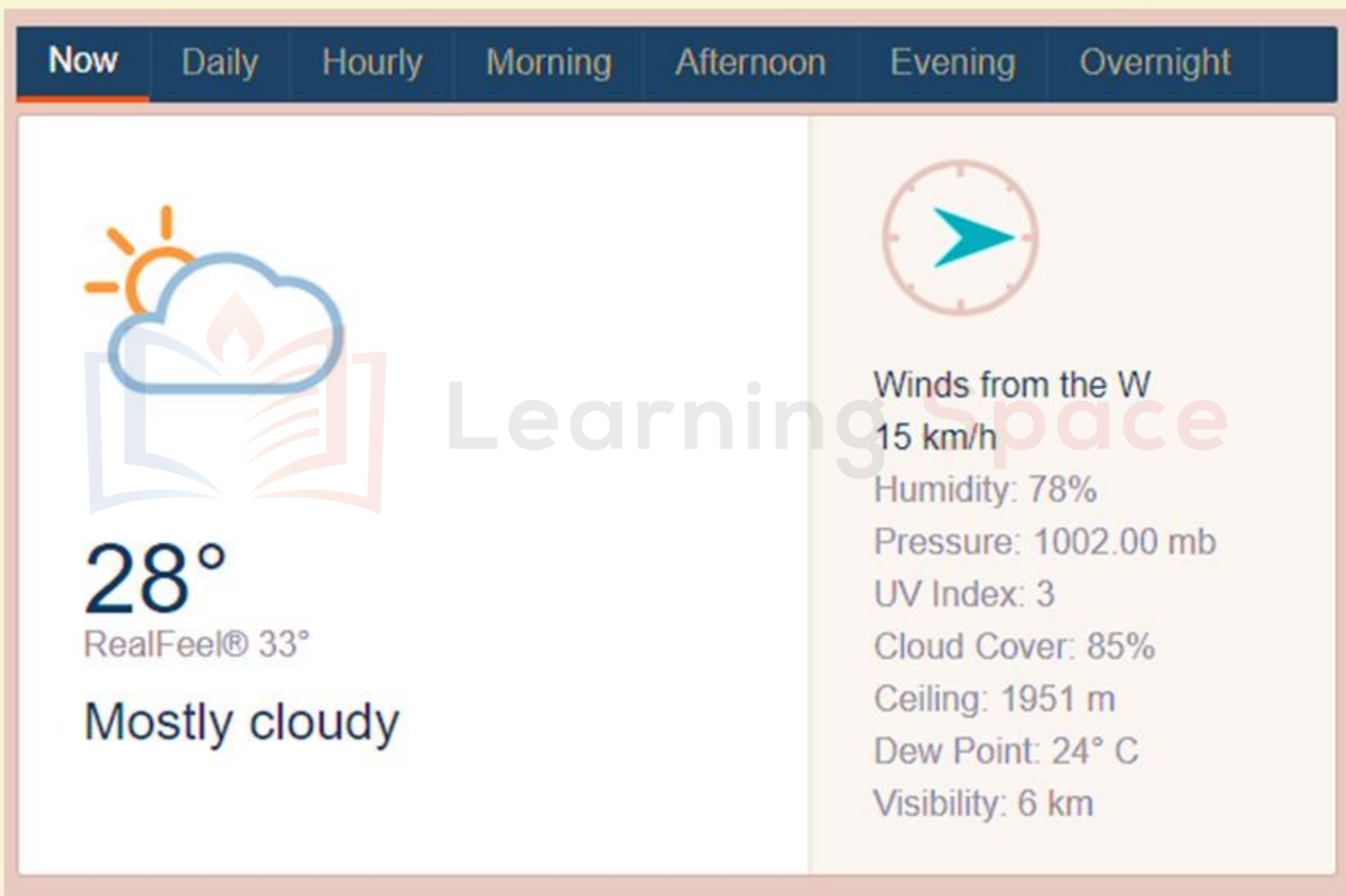
Precipitation may occur in different forms. Rainfall, sleet, snowfall and hail are the common forms of precipitation.

1. RAINFALL



- 1** The most common form of precipitation is rain and it is formed when **many cloud droplets coalesce into drops too large to remain suspended in the air.**
- 2** Rainfall occurs when the **dew point of air is above the freezing point** so that the condensing moisture is converted to water droplets.

LET'S LOOK AT THE REAL-TIME WEATHER DATA OF VIJAYAWADA



PRECIPITATION

FORMS OF PRECIPITATION

2. SNOWFALL



- 1 Snow consist of **masses of fine ice crystals**.
- 2 The dew point of air for this to occur should be **below the freezing point**.
- 3 This is likely to occur when the air contains very little moisture and its temperature falls below the freezing point before it attains its dew point.

PRECIPITATION

FORMS OF PRECIPITATION

3. HAIL

1 Hail consists of masses of ice with a layered structure. It occurs when there are very strong updrafts in the clouds carrying raindrops up to high altitude causing them to freeze.

2 This is common when the condensation occurs in a strong convective cell of air.



3 As the moisture contained in the air condenses, the strong updraft of air throws the raindrops upwards and they get frozen.

4 This process of upward thrust of rain drops may be repeated several times and a new layer of moisture will be added to the original drop of water with each such upward movement.

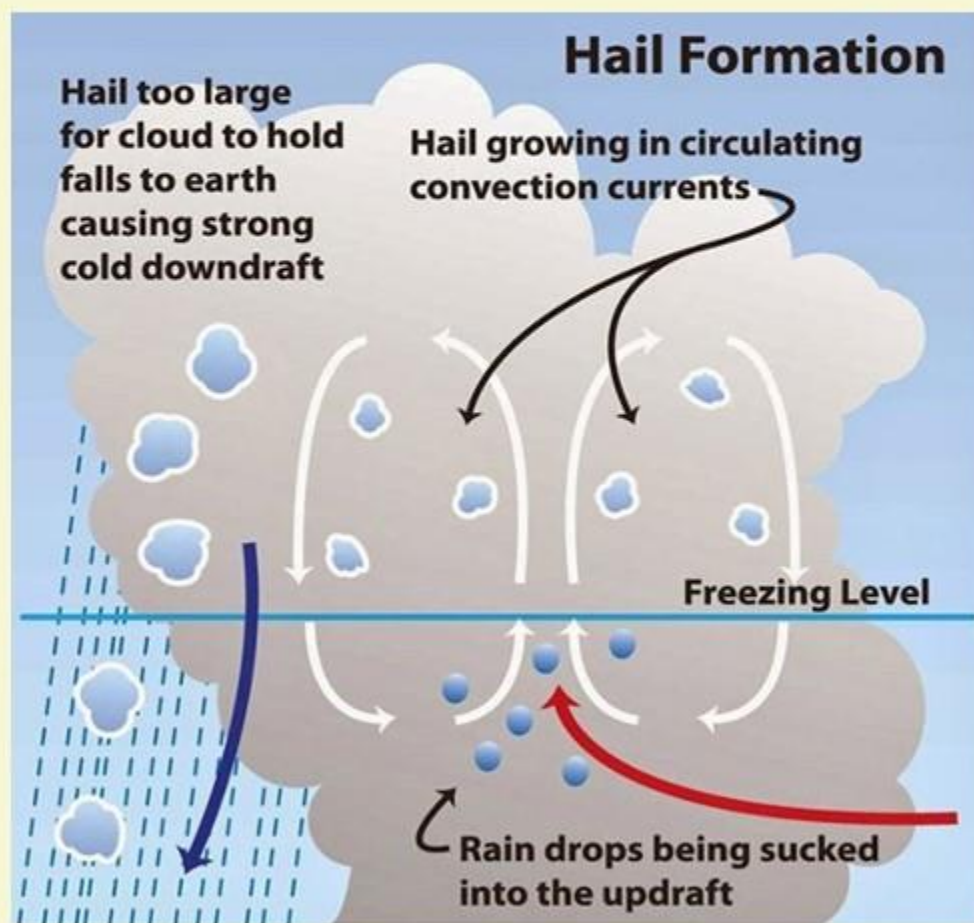
5 This gives a layered structure to the lump of ice and when its size becomes too large to be sustained in the updraft of air it ultimately falls to the ground

WHAT IS THE EXACT DIFFERENCE BETWEEN SNOW AND HAIL THEN?



- The difference between snow and hail is in the wind and temperature of the air when the snowflake or hail stone fall to the ground.
- Both hail and snow begin their journey in the same form, but something happens on the way down to create the difference between them.
- Snow is created when a droplet of water is captured on a dust particle. Since the air at a high level is very cold, the droplet immediately freezes and starts to crystalize. The crystal forms in the shape of a hexagon and forms a snow flake.

WHAT IS THE EXACT DIFFERENCE BETWEEN SNOW AND HAIL THEN?



- Hail also begins as a droplet of water. But instead of falling straight down, the wind (**strong updraft in intense thunderstorm clouds**) blows it higher causing it to accumulate additional droplets that freeze around it.
- The lump of ice that is created is heavier and falls down faster, and then the wind strikes it back up again. This causes it to gain more and more ice layers until finally it is too heavy to rise again, and it falls to the ground as hail stones.

PRECIPITATION

FORMS OF PRECIPITATION

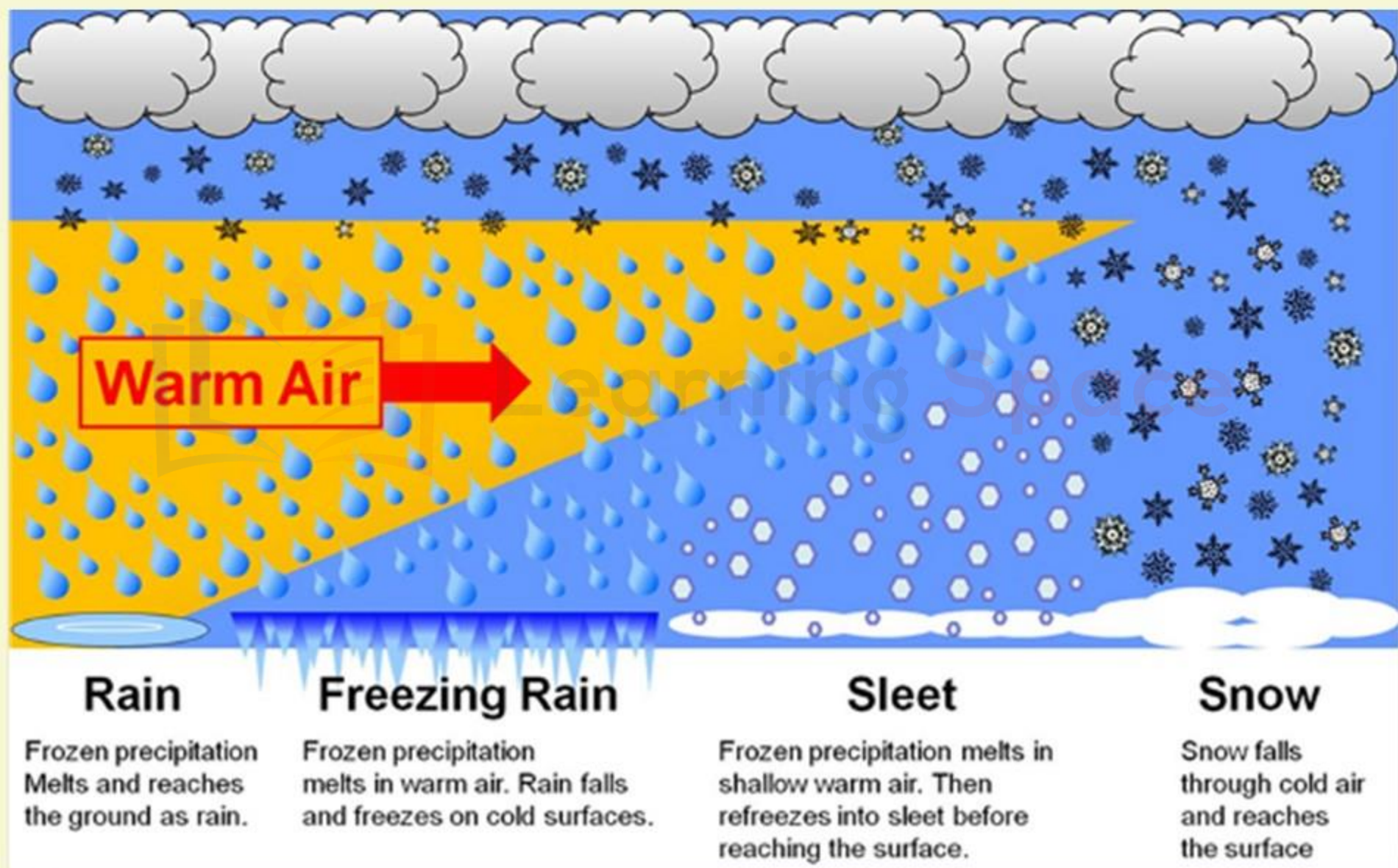


4. SLEET

- 1 Sometimes the **raindrops freeze before reaching the ground** and precipitation occurs in the form of ice pellets called Sleet.
- 2 This may happen when the **condensation is taking place in a relatively warmer air** and there is a layer of cold air beneath it and the raindrops passing through the colder layer are cooled below the freezing point.

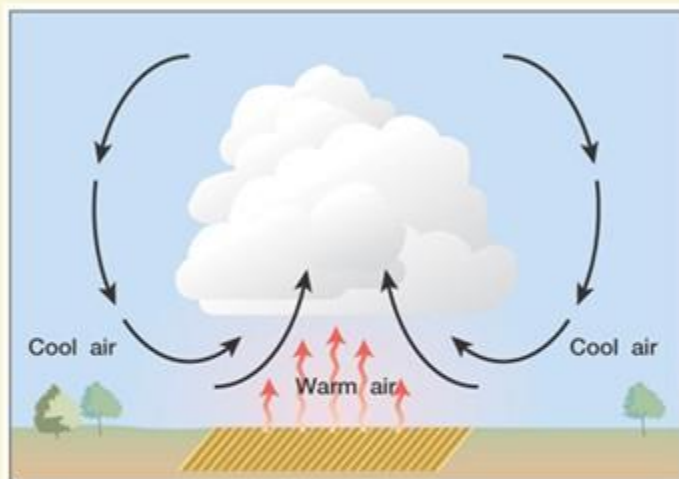
PRECIPITATION

FORMS OF PRECIPITATION

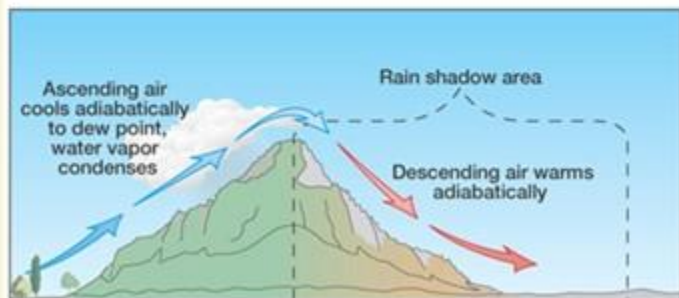


PRECIPITATION

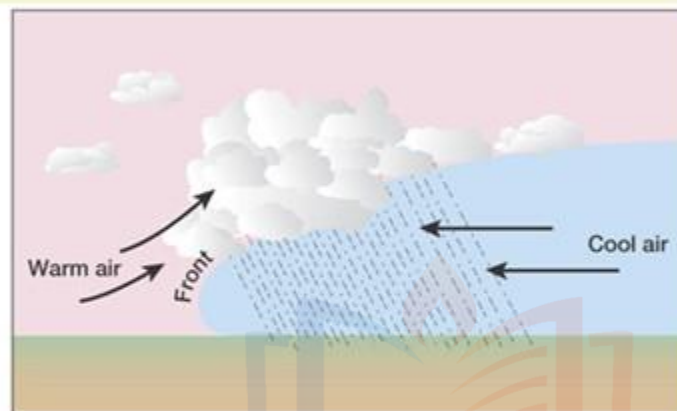
TYPES OF PRECIPITATION



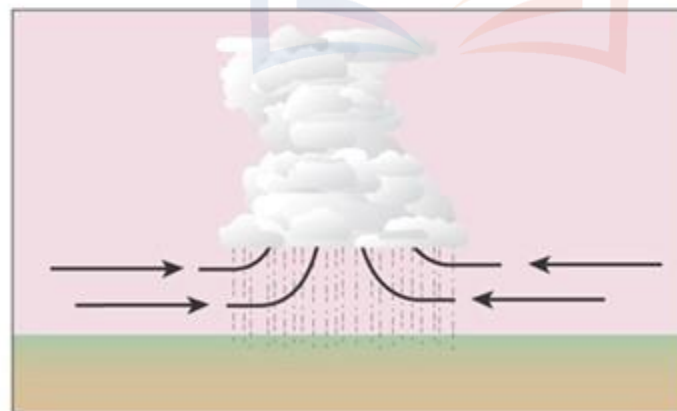
(a) Convectional



(b) Orographic



(c) Frontal



(d) Convergent

1 Although all forms of precipitation occur due to condensation of moisture in the air due to cooling of air through its uplift, the mechanism of uplift may vary from one instance to the other.

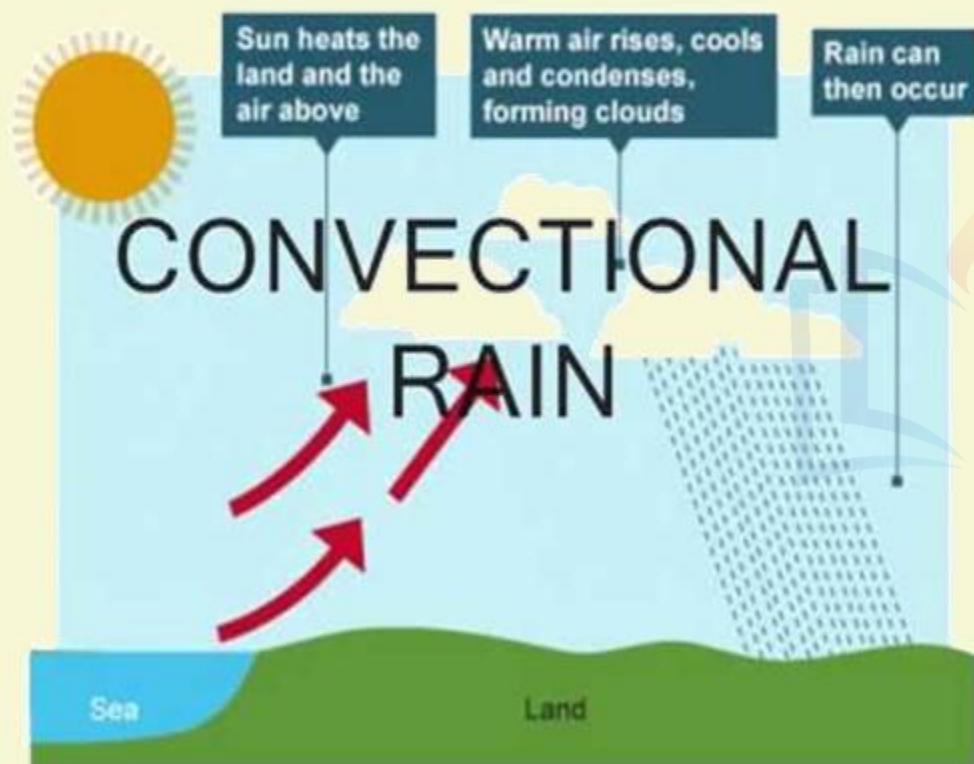
2 On the basis of the mechanism of uplift of air the precipitation can be

1. Convectional,
2. Orographic,
3. Cyclonic or Frontal.

PRECIPITATION

TYPES OF PRECIPITATION

1 CONVECTIONAL PRECIPITATION

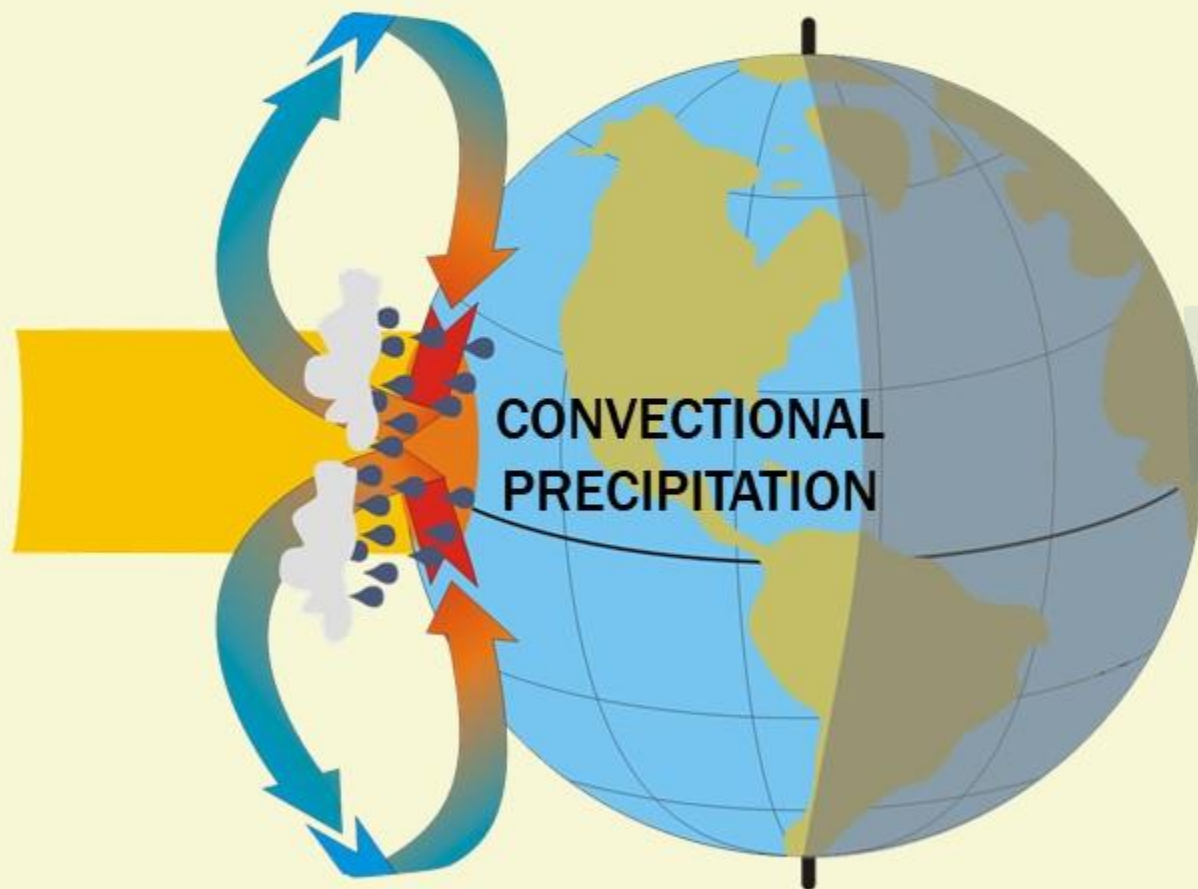


- 1 As the name suggests convectional precipitation is a **result of convection**.
- 2 Convection is defined as **a vertical movement of particles** and here it refers to vertical uplift of air due to **local instability**.
- 3 A parcel of air, it is warmer than the air in the surrounding areas has a tendency to rise upwards and such an air parcel rises as a convective cell.

PRECIPITATION

TYPES OF PRECIPITATION

1 CONVECTIONAL PRECIPITATION



4 This type of uplift of air is a common phenomenon in tropical regions, especially during the summer season.

5 Intense local heating of air may lead to convectional uplift of air and the air thus rising will be cooling adiabatically.

6 If the cooling due to uplift of air is sufficient, it may lead to condensation and precipitation. This type of precipitation is called convectional precipitation.

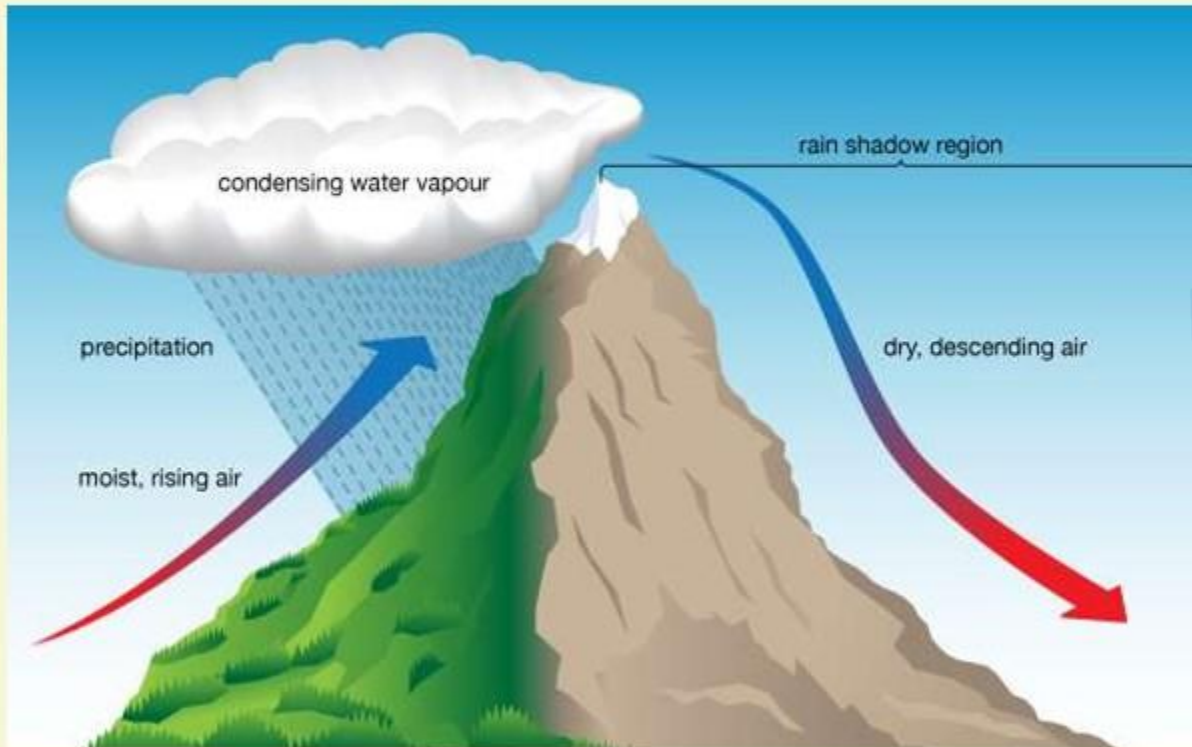
7 Much of the precipitation in Equatorial and Savanna regions occurs in this manner

PRECIPITATION

TYPES OF PRECIPITATION

2

OROGRAPHIC PRECIPITATION



1

Orographic precipitation is caused by landforms. It is sometimes also called mountain precipitation. This type the precipitation is caused by the air that is forced to rise up by physical obstructions like mountains.

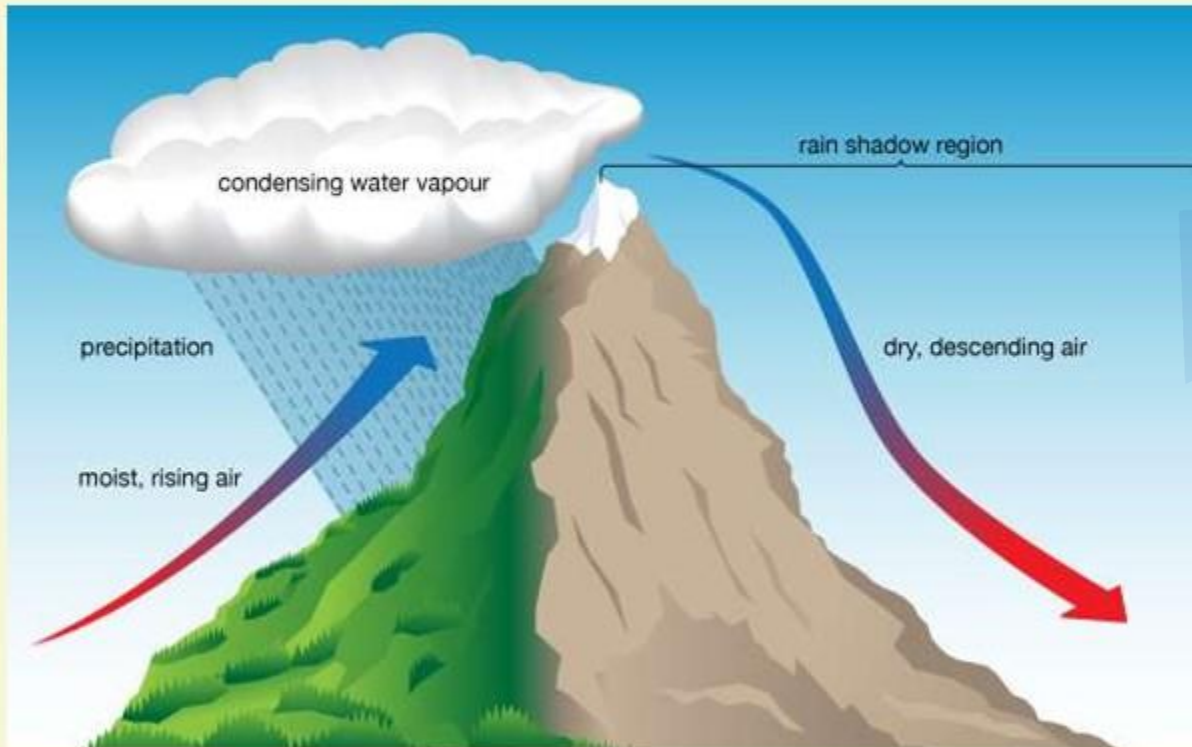
2

As the air rises to cross over such a barrier it is cooled adiabatically and condensation and precipitation occur if there is sufficient moisture in the air.

PRECIPITATION

TYPES OF PRECIPITATION

2 OROGRAPHIC PRECIPITATION



- 3** Much of the precipitation in the northern plains of India during the southwest monsoon season and most of the precipitation in the northeast India is of this type.

CONDENSATION TYPES

MOCK QUESTIONS

Q 8 What are forms of condensation? Describe the process of dew and frost formation.
(10 Marks) (150 words)

HINTS

Forms of condensation like rainfall, snowfall, hailstones, sleet, fog, mist etc.

For notes, please refer Module 26 and this Module.

DEW AND FROST



- 1 At night, the Earth surface and various objects such as leaves etc. may frequently become cooler than the air above it.
- 2 This causes water vapour contained in the air to **get condensed and deposit on the cooled surface**. This is known as dew.
- 3 The ideal conditions for its formation are clear sky, calm air, high relative humidity and cold and long nights.
- 4 On cloudy nights cooling of the surface is not complete **due to the greenhouse effect of the clouds** and dew is not likely to be formed.

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